

12. The formula of an unknown mineral is $\text{Mg}_a\text{X}_b(\text{CO}_3)_c(\text{OH})_d \cdot e\text{H}_2\text{O}$.

A student performs a series of experiments to calculate the values **a**, **b**, **c**, **d** and **e** and name element X.

Experiment Number	Experiment	Result
1	Heating a sample of 0.0200 mol of the mineral to dryness.	The sample loses 1.44 g of mass.
2	Addition of 0.0200 mol of the mineral to excess hydrochloric acid.	A volume of 490 cm^3 of carbon dioxide gas is released at 298 K and 1 atm pressure. The remaining solution has a green colour.
3	Addition of excess aqueous sodium hydroxide gradually to the solution produced in experiment 2.	A precipitate forms that shows a mixture of white and grey-green colours. Addition of excess sodium hydroxide leaves a white precipitate in a green solution.
4	Addition of a sample of 1.00×10^{-3} mol of the mineral to 25.0 cm^3 of HCl of concentration 1.00 mol dm^{-3} . Titration of the remaining acid against standard aqueous sodium hydroxide of concentration 0.500 mol dm^{-3} .	A volume of 14.00 cm^3 of the aqueous sodium hydroxide is required for complete reaction.

(a) Find the value of **e**, the number of water molecules in each formula unit. [2]

e =

(b) Find the value of **c**, the number of carbonate ions in each formula unit. [2]

c =



- (c) Explain the observations seen in experiment 3. Use these to name element X. [3]

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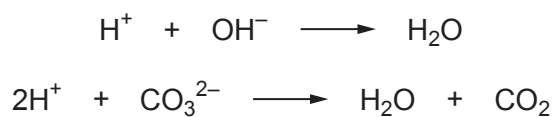
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- (d) In experiment 4 the acid reacts with all the hydroxide and carbonate ions.



Calculate the value of **d**, the number of hydroxide ions present in each formula unit. [4]

d =



- (e) The compound has 56 atoms in each formula unit.
Use your answers to (a), (b), (c) and (d), and the total number of atoms to find the formula of the mineral.

[3]

Examiner
only

Formula

END OF PAPER

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